# CHAPTER THREE

# DESIGN AND METHODOLOGY

## 3.1 THE PROPOSED SYSTEM

The proposed system is an **intelligent web-based agricultural market platform** designed to improve market access, reduce post-harvest losses, and enhance decision-making for smallholder farmers dealing with perishable food crops. The system integrates artificial intelligence techniques with digital marketplace functionalities to create a data-driven environment for agricultural trade.

Agricultural markets in Ghana and many developing economies are characterized by inefficiencies such as weak coordination between supply and demand, limited access to reliable market information, and high levels of post-harvest losses. These challenges are particularly severe for perishable crops such as fruits and vegetables, which have limited shelf lives and require timely market access. Traditional marketing systems depend heavily on intermediaries and informal trading practices, which often result in price distortions and reduced income for farmers. Although digital agricultural platforms have been introduced to address some of these challenges, many existing systems lack intelligent capabilities that can support predictive decision-making and product quality assessment.

To address these limitations, the proposed system introduces an intelligent platform that combines **digital agricultural trading with AI-driven prediction models**. The system allows farmers to input key agricultural data such as crop type and planting date. Based on this information, the system predicts the expected harvest date, estimates the freshness window of the produce, and determines the likely expiry period. These predictions enable farmers to plan harvesting and marketing activities more effectively.

In addition to freshness prediction, the system incorporates a **market demand analysis module** that evaluates current and historical market trends to identify areas where specific agricultural products are most needed. This functionality is particularly important for reducing post-harvest losses, as it allows farmers to redirect produce to high-demand locations when spoilage risk increases. By combining freshness prediction with demand intelligence, the system provides a comprehensive decision-support tool for farmers.

The system also integrates **trust and transparency mechanisms** to enhance user confidence. These include seller rating systems, feedback mechanisms, and AI-generated freshness scores. These features help buyers make informed purchasing decisions and improve the credibility of transactions conducted through the platform.

Overall, the proposed system transforms the traditional agricultural marketplace into an intelligent, data-driven ecosystem that supports efficient trade, reduces waste, and improves income stability for farmers.

## 3.1 SYSTEM SPECIFICATION

The system specification outlines the technical and operational requirements necessary for the implementation of the intelligent agricultural marketplace. The system is designed as a web-based platform supported by artificial intelligence models, database systems, and cloud-based services.

The architecture of the system integrates multiple components, including user interfaces, backend services, data processing modules, and machine learning models. These components work together to ensure that the system can collect data, process information, generate predictions, and deliver actionable insights to users in real time.

## HARDWARE REQUIREMENTS

The hardware components represent the physical devices required for accessing the system and supporting its operation. Since the system is web-based, the hardware requirements are minimal and focus mainly on user access and server infrastructure.

• **Smartphones**
Smartphones serve as the primary access device for most users, especially farmers. They are used to input crop data, view market demand information, and perform transactions on the platform. The system is optimized to run efficiently on low-cost smartphones to ensure accessibility.

• **Personal Computers (PCs) / Laptops**
Personal computers and laptops provide an alternative interface for accessing the system, particularly for buyers and administrators. These devices offer a larger display and improved processing capability, making them suitable for managing data and monitoring system activities.

• **Web Server (Cloud Server)**
The web server hosts the application, database, and backend services of the system. It is responsible for processing user requests, storing data, and running system operations. A cloud-based server is preferred to ensure scalability and reliability.

• **Internet Network Equipment (Router/Modem)**
Network devices such as routers and modems are required to provide internet connectivity for

both users and the server. These devices enable communication between system components and support real-time data exchange.

## SOFTWARE REQUIREMENTS

The software components represent the programs and platforms required for developing, running, and managing the proposed system. These tools support user interaction, backend processing, data storage, and secure transaction handling within the platform.

• **React**
React is used to develop the frontend of the system. It provides a responsive and interactive user interface that allows users to input data, view product listings, and access system features efficiently.

• **Node.js**
Node.js is used for backend development. It handles server-side operations, processes user requests, and manages communication between the frontend and the database.

• **Python**
Python is used for implementing the system's data processing and artificial intelligence logic. It supports the execution of prediction models and analysis of agricultural data.

• **Supabase**
Supabase is used as the database management system. It stores user information, crop data, and transaction records while providing real-time data access.

• **Paystack**
Paystack is used for processing payments within the system. It enables secure and efficient financial transactions between users.

• **Web Browser (Chrome, Edge)**
Web browsers are required to access the system. They provide the interface through which users interact with the platform without the need for installing additional applications.

## ARTIFICIAL INTELLIGENCE COMPONENTS

The artificial intelligence components form the core of the system's intelligent functionality. These components are responsible for analyzing agricultural data, generating predictions, and supporting decision-making processes for farmers and buyers.

• **Freshness Prediction Model (Regression Model)**
The freshness prediction model is used to estimate the harvest date, freshness duration, and expiry period of agricultural produce. It applies regression techniques to analyze input data such as crop type and planting date. This enables farmers to determine the optimal time to harvest and sell their produce, reducing post-harvest losses.

• **Market Demand Prediction Model (Time-Series Model)**
The market demand prediction model is used to forecast demand for agricultural products across different locations. It analyzes historical and real-time data using time-series techniques to identify demand trends. This allows the system to recommend markets where produce can be sold quickly, especially when it is nearing spoilage.

• **YOLOv5 (Computer Vision Model)**
YOLOv5 is used for image-based analysis of agricultural produce. It detects visual features such as color changes, texture, and defects that indicate freshness and quality. This enhances the system by providing visual verification alongside predictive models.

• **Machine Learning Libraries**
Machine learning libraries are used to develop, train, and implement the artificial intelligence models within the system. These libraries support data processing, model training, and prediction generation.

**Scikit-learn** is used for implementing regression algorithms and basic predictive models, particularly for estimating harvest dates and freshness duration. It provides efficient tools for model training and evaluation.

**TensorFlow** is used for advanced machine learning tasks such as deep learning and time-series forecasting. It supports the development of complex models used in market demand prediction.

**Pandas** is used for data manipulation and preprocessing. It enables the system to organize, clean, and structure agricultural data before it is used for model training.

**NumPy** is used for numerical computations and mathematical operations. It supports efficient handling of large datasets and improves the performance of machine learning algorithms.

**PyTorch** is used as the underlying framework for implementing and running deep learning models, particularly for the YOLOv5. It provides flexibility and efficiency in model development.

**OpenCV** is used for image processing tasks such as resizing, filtering, and enhancing images before they are analyzed by the computer vision model.

**Matplotlib** is used for data visualization. It helps display prediction results, trends, and analysis outputs in graphical form, making it easier for users to understand system insights.

## DEPLOYMENT

Deployment refers to the process of making the system available for use by end users. The proposed system is deployed using a cloud-based infrastructure, which ensures accessibility, scalability, and reliability.

The frontend of the system is deployed on a web hosting platform, allowing users to access the application through a web browser. The backend services are hosted on a cloud server, where they handle data processing, business logic, and communication with the database.

The database is managed using Supabase, which provides secure data storage and real-time synchronization. Payment integration is handled through Paystack, ensuring secure and efficient financial transactions.

Cloud deployment ensures that the system can scale as the number of users increases. It also provides high availability, allowing users to access the platform at any time. This approach reduces the need for physical infrastructure and simplifies system maintenance.

### 3.1.1 FUNCTIONAL REQUIREMENTS

The functional requirements define the specific operations and services that the proposed intelligent agricultural market system must perform. These requirements describe how the system interacts with users and how it processes data to achieve its objectives.

The system must provide user registration and authentication functionality to allow farmers and buyers to create accounts and securely access the platform. This ensures that all users are properly identified and that their activities can be managed within the system. After successful authentication, users should be able to access features relevant to their roles.

The system must allow farmers to input agricultural data, including crop type, planting date, and quantity of produce. This information is essential for generating predictions related to harvest time, freshness duration, and expiry periods. The system must process this data using artificial intelligence models to provide accurate and useful insights.

The system must support product listing functionality, enabling farmers to upload and display their produce on the platform. Each listing should include details such as product name, price, quantity, location, and freshness status. Buyers must be able to browse available products, search for specific items, and filter results based on their preferences.

The system must include a market demand analysis feature that evaluates data to identify areas where specific agricultural products are in high demand. This functionality is particularly important when produce is nearing spoilage, as it allows the system to recommend the best locations for selling such products quickly.

The system must also integrate a secure payment mechanism that allows buyers to purchase products directly through the platform. Payment processing should be handled efficiently, and both farmers and buyers should receive confirmation of transactions.

Additionally, the system must provide a feedback and rating feature that allows users to evaluate each other after transactions. This helps build trust within the platform and improves the reliability of user interactions. The system should also include administrative functions for monitoring system activities and managing users.

### 3.1.2 NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements describe the quality attributes and performance characteristics that the system must possess to operate effectively. These requirements focus on how the system performs rather than what it does.

The system must ensure high performance by processing user requests and generating responses within a short time. Features such as freshness prediction and market demand analysis should operate efficiently to provide real-time insights to users.

The system must be reliable and available at all times to support continuous access by users. Since agricultural transactions may occur at any time, the platform should minimize downtime and be capable of handling unexpected failures without significant disruption.

Security is a critical requirement of the system. The platform must protect user data and financial transactions through secure authentication mechanisms, data encryption, and controlled access. This ensures that sensitive information is not exposed to unauthorized users.

The system must be scalable to accommodate an increasing number of users and data over time. As more farmers and buyers join the platform, the system should maintain its performance without degradation.

Usability is also important, especially for farmers who may have limited experience with digital systems. The interface should be simple, intuitive, and easy to navigate, allowing users to perform tasks without difficulty.

Finally, the system must be maintainable and flexible, allowing developers to update features and fix issues easily. This ensures that the system can adapt to changing requirements and technological advancements over time.

## 3.2 SELECTION OF TECHNOLOGIES AND TOOLS

The selection of appropriate technologies and tools is essential for the successful development and implementation of the proposed intelligent agricultural market system. The choice of technologies determines the system's performance, scalability, security, and ability to integrate intelligent functionalities such as prediction and data analysis. In this study, modern web technologies, cloud-based services, and artificial intelligence tools are carefully selected to ensure that the system meets both functional and non-functional requirements.

The system adopts a web-based architecture, which requires a combination of frontend and backend technologies to enable user interaction and data processing. The frontend of the system is developed using React, a JavaScript library that supports the creation of responsive and interactive user interfaces. React allows the system to dynamically update content without reloading the entire page, thereby improving user experience. Its component-based structure also enhances code reusability and simplifies system maintenance.

On the server side, Node.js is used as the backend technology to handle application logic and communication between system components. Node.js is particularly suitable for real-time applications because of its asynchronous and non-blocking architecture, which enables the system to handle multiple user requests efficiently. This ensures that the platform remains responsive even when accessed by many users simultaneously.

The system uses Python as the primary language for implementing artificial intelligence functionalities. Python provides extensive support for machine learning and data analysis, making it suitable for developing predictive models such as freshness estimation and market demand forecasting. The integration of Python with backend services enables seamless execution of AI algorithms within the system.

For data storage and management, the system utilizes Supabase, which provides a cloud-based PostgreSQL database. Supabase offers real-time data synchronization, secure authentication, and easy integration with web applications. This ensures that user data, agricultural information, and transaction records are stored securely and can be accessed efficiently.

To support intelligent decision-making, the system incorporates machine learning libraries such as Scikit-learn, TensorFlow, Pandas, NumPy, PyTorch, and OpenCV. These tools enable the development and deployment of predictive models and computer vision algorithms. In particular, the YOLOv5 is used for image-based analysis of agricultural produce, providing visual assessment of freshness and quality.

The system also integrates Paystack as a payment gateway to facilitate secure financial transactions between farmers and buyers. Paystack supports multiple payment methods and ensures that transactions are processed efficiently and securely within the platform.

Additionally, web browsers such as Google Chrome and Microsoft Edge are used as the access interface for users. These browsers allow users to interact with the system without requiring additional software installation, thereby improving accessibility and ease of use.

Overall, the selected technologies and tools provide a robust foundation for the development of the intelligent agricultural marketplace. They ensure that the system is scalable, efficient, secure, and capable of supporting advanced artificial intelligence functionalities required to enhance agricultural market operations.

## 3.3 ARCHITECTURE OF THE SYSTEM

The architecture of the proposed intelligent agricultural market system defines the structural design and interaction between its various components. It describes how data flows through the system and how different modules work together to provide the required functionalities. The system adopts a **client-server architecture**, which separates the user interface from the data processing and storage components, ensuring efficiency, scalability, and maintainability.

The system is composed of four major layers: the presentation layer, application layer, data layer, and artificial intelligence layer. These layers work together to ensure seamless operation of the platform while maintaining clear separation of responsibilities. This layered approach improves system organization and allows each component to function independently while still contributing to the overall system performance.

The presentation layer represents the frontend of the system and is developed using React. This layer provides the interface through which users interact with the system. Farmers use this interface to input crop data, upload product details, and view AI-generated predictions, while buyers use it to browse products, view freshness indicators, and make purchases. The interface is designed to be responsive and user-friendly, ensuring accessibility across different devices such as smartphones and computers.

The application layer is implemented using Node.js and serves as the core processing unit of the system. It handles all business logic, including user authentication, product management, and communication with external services. This layer processes user inputs, interacts with the database, and coordinates the execution of artificial intelligence models. It also manages API requests and ensures that data is transmitted efficiently between the frontend and backend components.

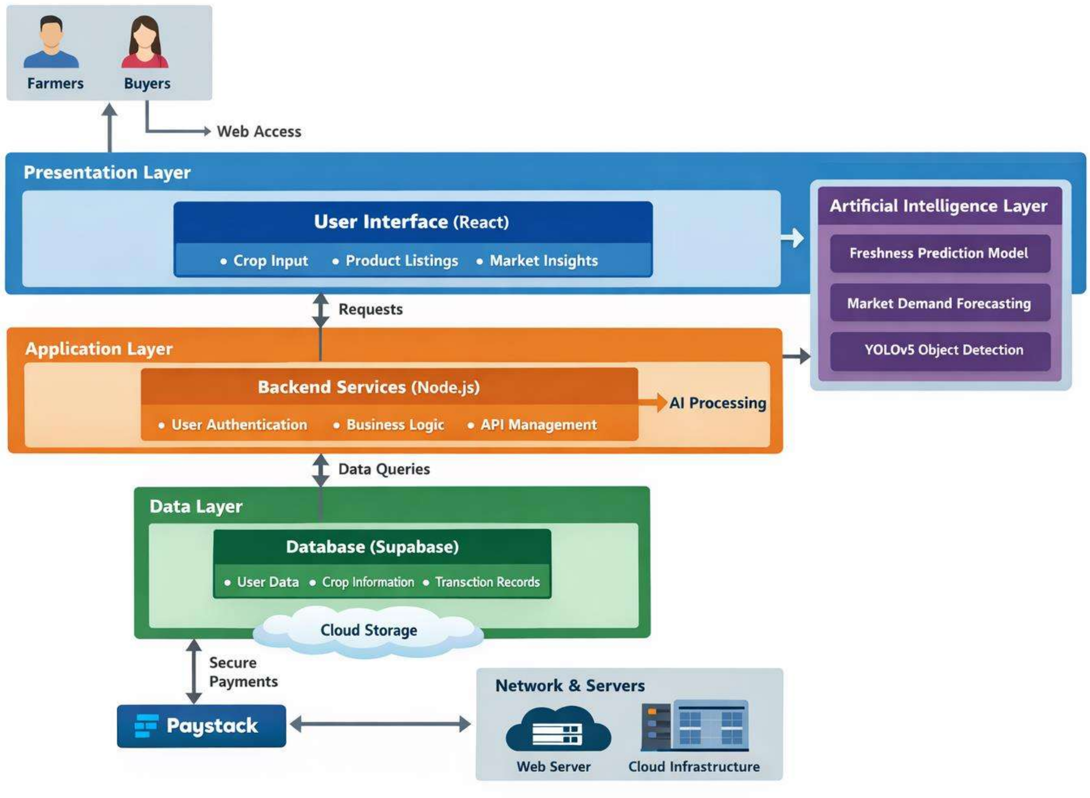
The data layer is responsible for storing and managing all system data. It is implemented using Supabase, which provides a secure and scalable cloud-based database solution. This layer stores user information, crop data, transaction records, and prediction outputs. The use of a centralized database ensures data consistency and enables real-time updates across the system.

The artificial intelligence layer integrates predictive models and computer vision algorithms into the system. This layer processes agricultural data to generate insights such as harvest time estimation, freshness prediction, and market demand forecasting. It also incorporates the YOLOv5 for analyzing images of agricultural produce to assess quality and freshness. The AI layer enhances the system's capability to support informed decision-making and improve efficiency in agricultural trading.

In addition to these layers, the system includes a payment integration component using Paystack, which facilitates secure financial transactions between users. Networking components ensure communication between the client and server, enabling seamless data exchange across the platform.

Overall, the system architecture is designed to be modular, scalable, and efficient. Each component performs a specific function while maintaining integration with other components. This design approach ensures that the system can be easily maintained, upgraded, and expanded to accommodate future technological advancements and increasing user demands.

## 3.4 DESIGN OF THE SYSTEM




The design of the proposed intelligent agricultural market system illustrates how the various system components interact to achieve the overall objectives of the platform. The system is structured using a layered approach, which enhances modularity, scalability, and ease of maintenance. The design integrates user interaction, backend processing, data management, and artificial intelligence functionalities into a unified framework.

As shown in the system design diagram, the system begins with users, specifically farmers and buyers, who access the platform through a web-based interface. Farmers interact with the system by entering crop-related data such as planting date, crop type, and quantity, as well as uploading images of produce where necessary. Buyers, on the other hand, interact with the system by browsing product listings, viewing freshness indicators, and making purchase decisions. This interaction occurs through the presentation layer, which provides a user-friendly interface developed using React.

The requests generated from the user interface are transmitted to the application layer, where backend services process the data. The backend, implemented using Node.js, handles core system functionalities such as user authentication, product management, and communication with the database. It also coordinates the execution of artificial intelligence models, ensuring that user inputs are analyzed and transformed into meaningful outputs such as freshness predictions and market demand recommendations.

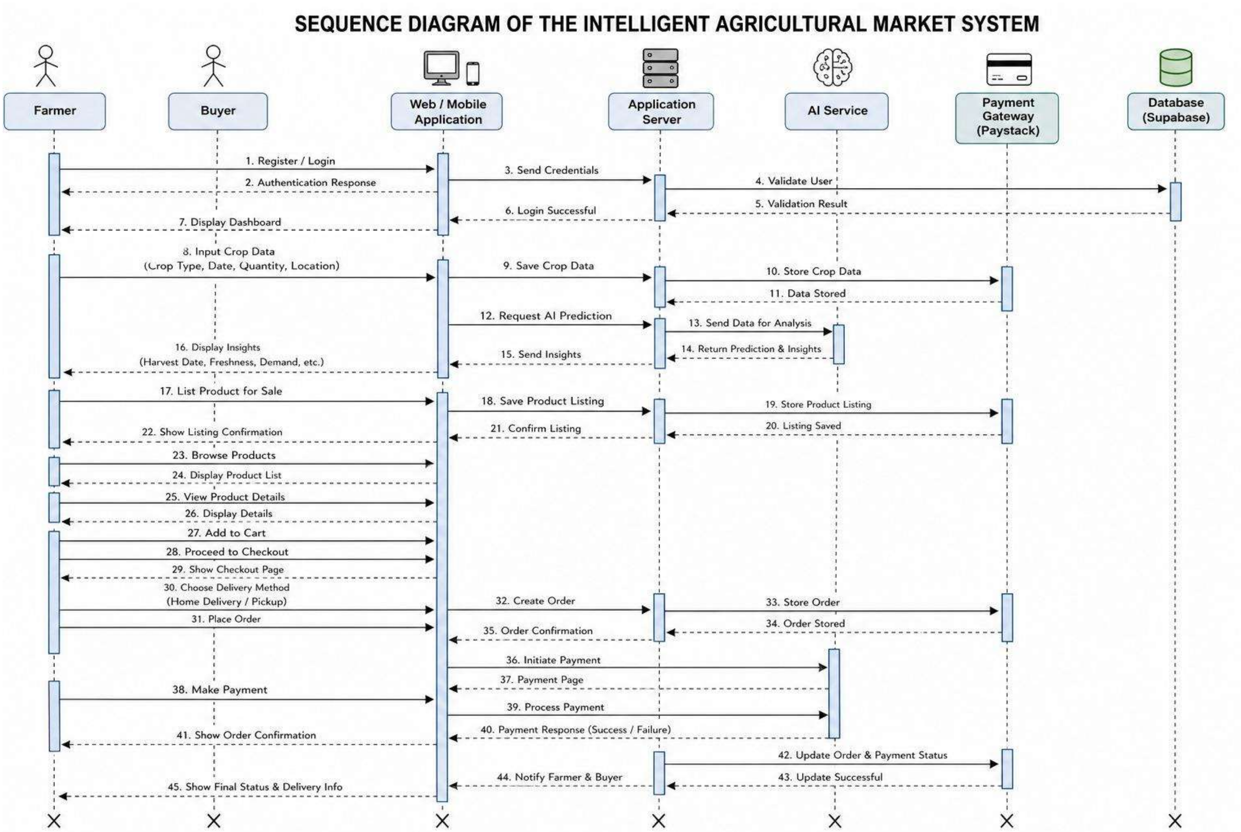
The system design also highlights the role of the data layer, which is responsible for storing and managing system data using Supabase. This includes user information, crop data, transaction records, and prediction results. The database supports real-time updates, ensuring that all users have access to accurate and up-to-date information.

A key component of the system design is the artificial intelligence layer, which enhances the system's decision-making capabilities. This layer processes agricultural data using predictive models to estimate harvest time, freshness duration, and expiry periods. It also includes the YOLOv5, which analyzes images of produce to assess visual quality and freshness. Additionally, market demand forecasting models analyze trends to recommend the best locations for selling produce, particularly when it is approaching spoilage.

The design further incorporates a payment processing component using Paystack, which enables secure transactions between buyers and farmers. Payment information is processed securely, and transaction confirmations are provided to both parties, ensuring transparency and trust within the system.

Finally, the system operates within a cloud-based environment that includes web servers and networking infrastructure. This ensures that the platform is accessible to users at all times and can scale to accommodate increasing demand. The integration of all these components creates a robust and efficient system capable of improving agricultural market operations through intelligent data-driven solutions.

### 3.4.1 SEQUENCE DIAGRAM OF THE PROPOSED SYSTEM




The sequence diagram of the proposed intelligent agricultural market system illustrates the interaction between the various system components over time. It shows how users, including farmers and buyers, communicate with the system through the user interface, and how requests are processed by the backend services, database, and artificial intelligence components.

The process begins when a user logs into the system through the web interface. The authentication request is sent to the backend server, which verifies the user credentials using the database. Upon successful authentication, the user is granted access to the system based on their role.

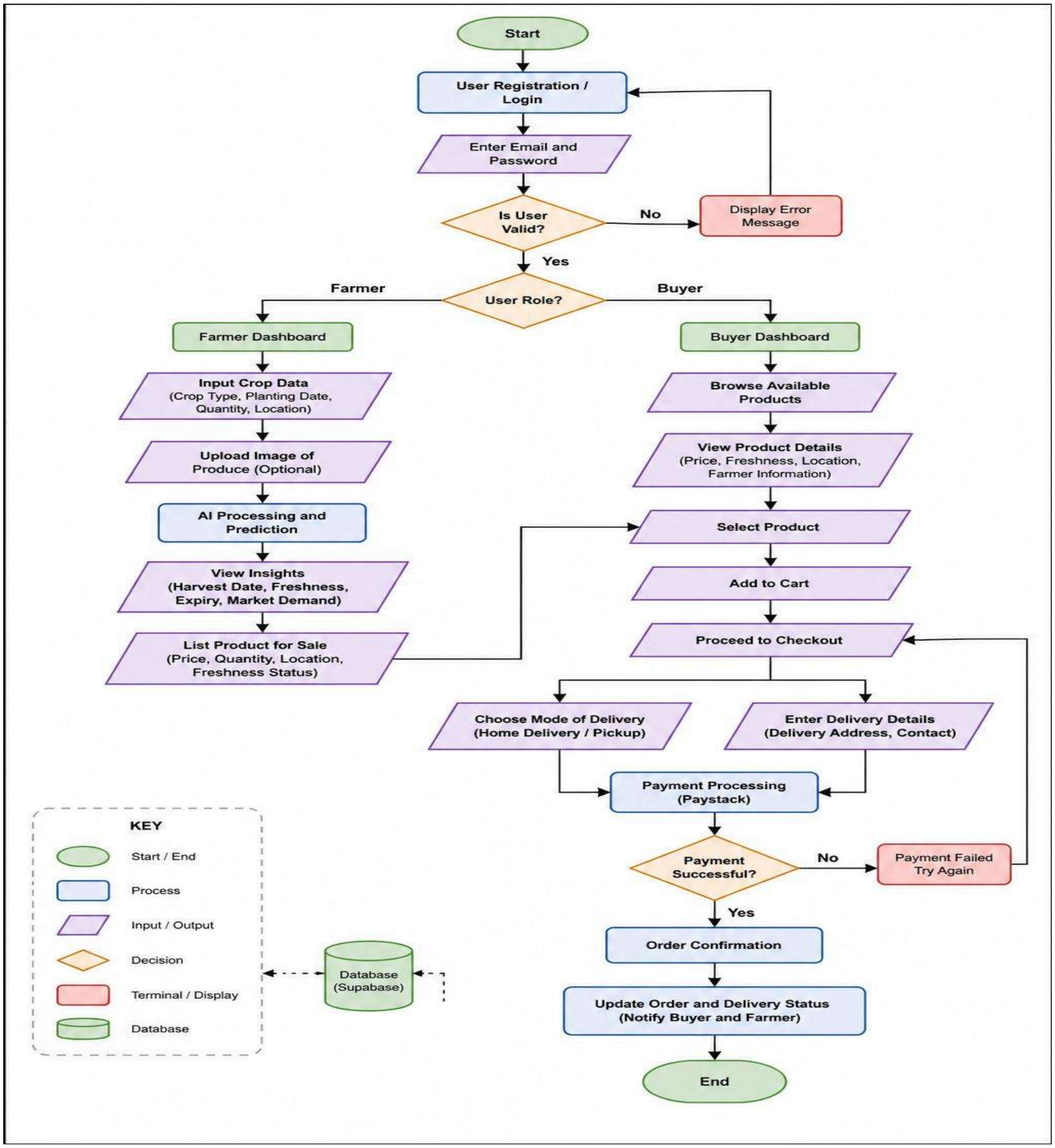
When a farmer inputs crop data such as crop type, planting date, and quantity, the information is transmitted to the backend and stored in the database using Supabase. The backend then forwards the data to the artificial intelligence components for analysis. The system uses predictive models to estimate harvest date, freshness duration, and expiry period. If images are provided, the YOLOv5 analyzes the images to assess produce quality.

The processed results are returned to the backend and displayed to the farmer through the user interface. The farmer can then list the product on the platform. Buyers interact with the system by browsing available listings, selecting products, and proceeding to checkout.

During checkout, the buyer selects a delivery method and confirms the order. Payment is processed securely through Paystack. Once the transaction is successful, the system updates the database and sends confirmation notifications to both the buyer and the farmer.

Overall, the sequence diagram demonstrates the step-by-step interaction between system components, highlighting how data flows through the system to support intelligent decision-making and efficient agricultural trading.

### 3.4.2 FLOWCHART OF THE SYSTEM




The flowchart of the proposed system presents a graphical representation of the operational workflow of the intelligent agricultural marketplace. It illustrates the sequence of processes involved in user interaction, data processing, artificial intelligence analysis, and transaction handling within the system.

The process begins with user login, where the system authenticates the user before granting access. After authentication, the system determines the user role and directs the user to the appropriate interface. Farmers are provided with options to input crop data, while buyers are directed to browse available products.

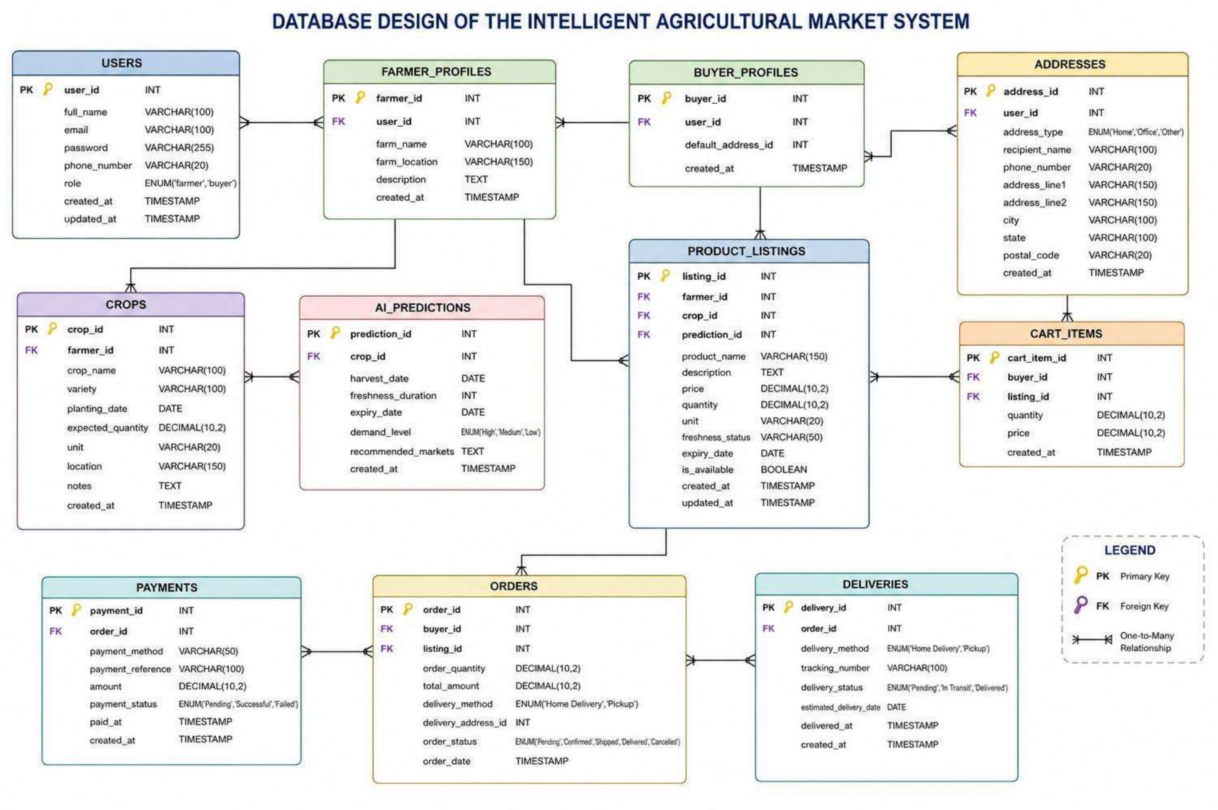
When a farmer enters crop information, the system processes the data using artificial intelligence models to generate predictions such as harvest date, freshness duration, and expiry period. The system also performs image-based analysis using YOLOv5 to evaluate produce quality. Based on these analyses, the system provides decision-support insights to the farmer.

The farmer then lists the product on the platform, making it available to buyers. Buyers browse product listings, select desired items, and proceed to the checkout stage. At checkout, the buyer chooses a delivery method and confirms the order.

The system processes payment securely using Paystack and updates transaction records accordingly. After successful payment, the system generates confirmation notifications and initiates the delivery or pickup process. The workflow ends after the transaction is completed.

This flowchart provides a clear and structured representation of the system's operations, showing how different processes are interconnected to achieve efficient agricultural market transactions.

### 3.4.3 DATABASE DESIGN




The database diagram of the proposed system illustrates the structure of data storage and the relationships between different entities within the intelligent agricultural marketplace. It provides a visual representation of how data is organized and managed to support system operations.

The system utilizes a relational database implemented using Supabase. The database consists of key entities, including Users, Crops, Listings, Transactions, and AI Predictions, each of which plays a critical role in the system.

The Users entity stores information about all system users, including farmers and buyers. The Crops entity contains agricultural data entered by farmers, such as crop type and planting date. The Listings entity represents products available for sale, linking farmers to buyers within the marketplace.

The Transactions entity records all purchase activities, including payment details, delivery method, and transaction status. This ensures that all financial operations are properly tracked and managed. The AI Predictions entity stores outputs generated by the artificial intelligence models, such as harvest date, freshness duration, and demand recommendations.

Relationships between these entities are established using primary and foreign keys. For example, the Users entity is linked to the Crops and Listings entities, while the Listings entity is connected to the Transactions entity. The Crops entity is also linked to the AI Predictions entity, enabling the system to associate prediction results with specific agricultural data.

Overall, the database design ensures efficient data organization, integrity, and accessibility. It supports seamless integration between user activities, artificial intelligence processing, and transaction management, enabling the system to operate effectively and provide intelligent agricultural market solutions.